

Adventist University of Central Africa

HIV & TB PATIENT MONITORING SYSTEM

Case Study: Dream Medical Center

A final year project Presented in partial fulfillment of the Requirements for the degree
of BACHELOR OF SCIENCE IN INFORMATION
TECHNOLOGY

Major in

Software Engineering

By

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ABSTRACT

A final year Project for Bachelor Degree of Science in Information Technology

Emphasis in Software Engineering

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TITLE: HIV & TB PATIENT MONITORING SYSTEM

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HIV and Tuberculosis (TB) remain major public health challenges in Rwanda, requiring strict treatment adherence, timely follow-up, and coordinated care between community and facility-level health services. Community Health Workers (CHWs) play a critical frontline role in Rwanda's last-mile healthcare delivery; however, continued reliance on paper-based documentation results in delayed reporting, transcription errors, fragmented patient records, and a complete absence of real-time visibility into whether patients are actually adhering to their treatment between CHW visits.

This project designs and implements an integrated intelligent HIV & TB Patient Monitoring System for Dream Medical Center, aimed at digitizing community-level care workflows, embedding artificial intelligence into adherence monitoring, and strengthening coordination between community and facility-level health services. The solution employs an offline-first Flutter mobile application, a Spring Boot cloud backend, a Python-based AI microservice, and Fast Healthcare Interoperability Resources (FHIR) standards to enable secure, interoperable, and real-time data exchange with hospital Electronic Health Record (EHR) systems.

Key system capabilities include AI-powered risk prediction, false confirmation detection, CHW prioritization, stock forecasting, and early warning detection for treatment failure clusters. Designed for low-connectivity rural settings, the platform uses user-centered design to support CHWs and patients with different digital literacy levels. Built with Agile methodology, it enables proactive HIV and TB care by predicting treatment interruptions early and guiding timely clinical responses. The project supports better patient outcomes and aligns with Rwanda's National Digital Health Strategic Plan 2020–2025

DECLARATION

I, **IGIHOZO Nelly**, a student at Adventist University of Central Africa in the Faculty of Information Technology, Department of Software Engineering, declare that this research project is my own original work. I wrote it myself based on my own knowledge and experience. This work has never been submitted to any university or college before, either in part or in full.

Signature:

Date:/...../.....

APPROVAL

I, **MANIRAHO Laurent** hereby certifies that I have supervised this report; has been done under my supervision and submitted with my approval.

Signature:

Date...../...../.....

DEDICATION

To my parents,

To my beloved sisters and brothers,

To my classmates, friends, and relatives,

To my supervisor for his guidance,

I dedicate this final report.

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LIST OF ABBREVIATIONS

API:	Application Programming Interface
ART:	Antiretroviral Therapy
AUCA:	Adventist University of Central Africa
CHW:	Community Health Worker
DB:	Database
DOT:	Directly Observed Therapy
HER:	Electronic Health Record
FHIR:	Fast Healthcare Interoperability Resources
HIV:	Human Immunodeficiency Virus
HL7:	Health Level Seven
IDE:	Integrated Development Environment
IT:	Information Technology
JWT:	JSON Web Token
mHealth:	Mobile Health
MOH:	Ministry of Health
OOP:	Object-Oriented Programming
REST:	Representational State Transfer
SMS:	Short Message Services
SMTP:	Simple Mail Transfer Protocol
TB:	Tuberculosis
UI:	User Interface
UML:	Unified Modeling Language
URL:	Uniform Resource Locator
UX:	User Experience
VOT:	Video-Observed Therapy

CHAPTER 1

GENERAL INTRODUCTION

Introduction

Rwanda has made significant and internationally recognized progress in the control of HIV and Tuberculosis (TB), achieving antiretroviral therapy (ART) coverage rates exceeding 90% and demonstrating sustained national commitment to public health improvement (Rwanda Ministry of Health, 2007). Through structured community health programs, targeted policy-driven interventions, and investments in health workforce capacity, the country has strengthened prevention, diagnosis, and treatment services across its national health system, positioning itself as a regional leader in HIV/TB management.

Despite these achievements, critical gaps remain at the community level particularly in what is commonly referred to as the 'last mile' of healthcare delivery. Community Health Workers (CHWs), who serve as the primary interface between households and health facilities, continue to rely heavily on paper-based documentation during home visits, medication follow-ups, and symptom monitoring. Although this manual system supports basic data collection in resource-limited settings, it creates significant and systemic delays in reporting, increases the risk of transcription errors, and fragments patient records across community and facility levels in ways that undermine care quality.

The growing global adoption of mobile health (mHealth) technologies and internationally recognized interoperability standards such as Fast Healthcare Interoperability Resources (FHIR) presents a concrete and timely opportunity to transform this landscape. By digitizing community-level workflows and enabling secure, standardized data exchange with national health systems, Rwanda can bridge the persistent gap between CHW activities and facility-based providers, strengthen inter-level coordination, and significantly enhance data-driven decision-making across the full HIV/TB care continuum. This project directly responds to that opportunity through the design and implementation of the HIV & TB Patient Monitoring System at Dream Medical Center.

Background of the Study

Rwanda's national HIV and TB programs are among the most successful in sub-Saharan Africa, supported by a national ART rollout coordinated through the Treatment and Research AIDS Centre and integrated TB-HIV collaborative activities formally established through national policy since 2005 (Rwanda Ministry of Health, 2007; Mutembayire et al., 2025). Rwanda has achieved viral suppression rates exceeding 90% among people on antiretroviral therapy and TB treatment success rates consistently above World Health Organization global targets. Central to this success is a national Community Health Worker (CHW) program maintaining approximately 45,000 CHWs organized in teams of three per village, providing health education, treatment adherence support, and Directly Observed Therapy for TB patients under health center supervision (CHW Central, 2018). These outcomes reflect sustained investments in health infrastructure, community health systems, and national health information management over more than two decades.

However, CHW documentation tools have not kept pace with growing program demands. Manual patient registers, visit forms, and medication tracking sheets create persistent challenges in data quality, timeliness, and integration with facility-level health information systems, challenges directly experienced at Dream Medical Center, the pilot site for this project. No existing tool gives clinical staff real-time visibility into whether patients are adhering to treatment between CHW visits. By the time deteriorating adherence is noticed, weeks of missed doses may have already occurred, increasing the risk of treatment interruption, drug resistance, and preventable morbidity.

The need for digital transformation at the community care level is therefore both urgent and well-evidenced. Beyond simple digitization, the scale and complexity of Rwanda's community HIV and TB program demands an intelligent system capable of predicting which patients are most at risk of treatment failure before that failure occurs and directing timely intervention to prevent it. This project was developed in direct response to those operational gaps and aligns with Rwanda's National Digital Health Strategic Plan 2020-2025, which prioritizes strengthening the interoperability of health information systems and improving service delivery through digital tools (Rwanda Ministry of Health, 2020).

Problem Statement

Despite Rwanda's significant progress in HIV and TB control, community-level patient monitoring at Dream Medical Center remains heavily dependent on fragmented, paper-based systems that undermine care coordination, data quality, and timely clinical intervention. The consequences of these systemic deficiencies extend beyond administrative inefficiency. They directly affect patient safety, treatment continuity, and the ability of health managers to oversee program performance with any degree of accuracy or timeliness. Most critically, the current system has no mechanism to predict which patients are approaching treatment failure before that failure occurs, meaning that clinical intervention is always reactive rather than preventive.

- **Delayed Data Transmission:** Community-collected patient data reaches health facilities between one and two weeks after the actual point of care, critically limiting the ability of providers to respond promptly to missed doses or emerging complications. In the context of HIV and TB treatment, where consistent daily adherence is essential to prevent drug resistance and treatment failure, a two-week reporting lag is clinically unacceptable. A patient who begins missing doses the day after a CHW visit may go entirely undetected until the following supervision cycle, by which point the treatment gap may have grown to a size that requires intensive and costly clinical intervention to reverse. The absence of any real-time data channel between the community and the facility means that deteriorating patient conditions are invisible to clinical staff until they become crises.
- **Paper-Based Documentation:** CHWs rely entirely on manual documentation during home visits, consuming disproportionate time, increasing transcription errors, and producing incomplete and inconsistent patient records. Field observations at Dream Medical Center revealed that CHWs spend an estimated 30 to 40 percent of each home visit on paperwork rather than direct patient interaction. Entries are frequently reconstructed from memory after the visit rather than recorded in real time, further reducing accuracy. Illegible handwriting, missing fields, and inconsistent patient identifiers compound these quality issues at every subsequent stage of data handling, making it impossible to build the reliable longitudinal patient record that meaningful adherence analysis requires.

- **Fragmented Information Systems:** The absence of any real-time integration between community and facility systems results in duplicated patient records, fragmented treatment histories, and non-functional referral confirmation pathways. A single patient's complete care record may be distributed across a CHW field register, a facility summary register, a clinician treatment file, and a pharmacy dispensing log, none of which are formally reconciled. This fragmentation means that clinical decisions during facility consultations are routinely made without access to recent community-level observations, adherence data, or CHW-reported side effects. Referrals initiated by CHWs have no digital confirmation pathway, meaning there is no systematic way for a CHW to know whether a patient they referred actually attended the facility appointment.
- **Unreliable Medication Stock Management:** CHW stock records are maintained manually and are frequently out of date, contributing to preventable medication stockouts that interrupt treatment continuity for vulnerable patients. There is no automated forecasting mechanism, no electronic visibility into CHW stock levels for facility pharmacists, and no alert system to flag approaching stockouts before they become critical. When a CHW exhausts their medication supply unexpectedly, patients may go without doses for several days while the CHW travels to the facility for resupply. For HIV patients on antiretroviral therapy and TB patients on fixed-duration regimens, even short interruptions in medication supply can have serious clinical consequences including the development of drug resistance.
- **Absence of Real-Time Analytics:** Program managers cannot identify adherence trends, monitor CHW performance, or make evidence-based decisions due to the lack of aggregated real-time data. Answering basic program management questions such as which patients are at highest risk of treatment interruption or how many patients have missed more than two consecutive doses this month requires days of manual data compilation that is rarely completed comprehensively. More critically, the current system has no predictive capability. It cannot identify a patient whose adherence is gradually declining before that decline becomes a treatment failure, meaning clinical intervention is always reactive rather than preventive.

- **Gap Between Policy and Implementation:** Although Rwanda has prioritized digital health transformation through its National Digital Health Strategic Plan 2020-2025, a significant gap remains between national policy intent and practical FHIR-compliant solutions connecting community care with facility systems. Few community-level digital tools currently deployed in Rwanda support HL7 FHIR R4 interoperability standards, meaning that even where mobile data collection exists it remains isolated from facility Electronic Health Record systems. Dream Medical Center exemplifies this gap. Despite operating within a nationally digitized health system, its community care layer remains entirely disconnected from any digital infrastructure, and no tool currently in use incorporates the artificial intelligence capabilities needed to transform raw adherence data into actionable clinical predictions.

Choice and Motivation of Study

The motivation for this study arose from direct observation of, and engagement with, the operational challenges of community-level HIV and TB care at Dream Medical Center, Rwanda. Through structured interaction with Community Health Workers, facility nurses, data management staff, and the facility administration, it became unmistakably clear that the existing paper-based patient monitoring process creates significant, measurable inefficiencies that negatively affect both patient safety and CHW effectiveness. These were not theoretical problems identified through literature review alone they were observed realities with documented consequences for real patients.

The choice to develop the HIV & TB Patient Monitoring System was driven by three interconnected and mutually reinforcing motivations. First, at the institutional level, the system directly addresses documented operational failures at Dream Medical Center: delayed data transmission, chronic incompleteness in patient records, unreliable adherence monitoring, and the absence of any functional mechanism for real-time communication between CHWs and facility clinical staff. Second, at the academic level, the project provided an opportunity to apply and integrate core competencies acquired throughout the Software Engineering program at AUCA, including systems analysis and design, relational database architecture, mobile application development, REST API design, and healthcare interoperability standards. Third, at the public health level, the project contributes to Rwanda's national digital health transformation agenda by

delivering a practically implemented, FHIR-compliant platform that connects the community and facility layers of the health system in a standards-based, scalable way.

Beyond these three primary motivations, the project also holds significance from a research perspective. There is a documented gap in the literature regarding practical implementations of FHIR standards in community health worker settings in low-resource countries. Most published evidence on FHIR adoption focuses on hospital and specialist settings in high-income countries. By implementing and documenting a FHIR-based community health monitoring system in Rwanda, this project contributes original practical knowledge to an underexplored domain.

Objectives of the Study

General Objective

To design and implement a comprehensive mobile-based HIV and TB patient monitoring system that facilitates secure, FHIR-compliant care coordination between Community Health Workers and facility-level providers at Dream Medical Center, Rwanda, thereby improving treatment adherence monitoring, data quality, and continuity of care for HIV and TB patients.

Specific Objectives

- To enable offline-capable mobile registration of HIV and TB patients with full demographic and clinical detail capture, multi-language support, and built-in duplicate detection mechanisms.
- To implement an automated medication adherence calculation engine that tracks dose records in real time, computes rolling adherence percentages, and generates timely missed-dose alerts to enable early CHW intervention.
- To develop a CHW-level medication inventory management module that records dispensing events, tracks remaining stock balances, generates low-stock alerts, and provides basic shortage forecasting to prevent treatment interruptions.
- To enable bidirectional, HL7 FHIR R4-compliant, and encrypted data exchange between the mobile application and hospital Electronic Health Record (EHR) systems, ensuring community data is accessible to facility clinicians in near real time.

- To provide visual analytics and performance reporting dashboards for supervisors and facility clinicians to monitor individual patient adherence trends, CHW activity levels, and program-level outcome indicators.

Scope of Study

Functional Scope

The system provides secure login and role-based access control so that each type of user sees only the functions relevant to their role. It supports patient registration and medical history management, capturing demographics, diagnosis details, treatment start dates, and household location with automatic duplicate detection. CHWs can record home visits using structured digital forms covering medication adherence, symptoms, side effects, and next visit scheduling. The system calculates and displays rolling adherence percentages automatically and generates alerts when a patient misses doses or falls below the configured adherence threshold. Medication stock is tracked at the CHW level with automated low-stock alerts and digital resupply requests. A web-based dashboard gives facility providers and supervisors real-time visibility into adherence trends, visit completion rates, and CHW performance. The mobile application operates fully offline using local storage and synchronizes automatically with the server when connectivity is restored, without requiring any action from the CHW.

Technical Scope

The mobile application is built with Flutter and runs on Android devices, using SQLite for local offline storage. The backend is built with Spring Boot and uses a PostgreSQL database hosted on a cloud server. The web dashboard is built with Next.js and runs on any modern web browser. A Python microservice handles the AI processing including adherence risk scoring, false confirmation detection, stock forecasting, and early warning detection. A HAPI FHIR server manages all data exchange with hospital Electronic Health Record systems using the HL7 FHIR R4 standard. All components are packaged using Docker for consistent and simple deployment.

Geographic Scope

The initial deployment and testing of the system takes place at Dream Medical Center in Rwanda. The system is designed to be expanded to additional health facilities and

community health programs without requiring changes to the core software. The cloud infrastructure can be hosted on any server with a reliable internet connection, including affordable options available within Rwanda.

Target Users

The system is designed for five types of users. Patients receive medication reminders and confirm doses through a simple mobile interface. Community Health Workers use the mobile application to register patients, record home visits, manage medication stock, and act on AI-generated priority lists. Facility providers use the web dashboard to monitor patient adherence, receive clinical alerts, and manage referrals. Supervisors use the web dashboard to monitor CHW performance and view population-level adherence trends. System administrators manage user accounts, configure system settings, and review audit logs.

Methodology and Techniques Used in the Study

The study adopted a practical approach that combined gathering information from people directly involved in HIV and TB community care, reviewing existing documentation, and building the system through a step-by-step development process. Each layer of the system, the Flutter mobile application, the Spring Boot backend, the Python AI microservice, and the web dashboard, was built and tested before the next layer was added, which made it easier to find and fix problems as the work progressed.

Interviews

An interview is a structured conversation used to gather information and understand the experiences and views of people who are directly involved in a subject (Wikipedia, 2024). Interviews were conducted with key stakeholders at Dream Medical Center including six Community Health Workers, three facility nurses, one data manager, and the facility medical director. The interviews were designed to understand how patient monitoring currently works, what problems are most commonly experienced, what features a digital system would need to be useful, and what concerns people might have about adopting new technology. The findings from these interviews directly shaped the requirements described in Chapter Two and the design decisions described in Chapter Three.

Observation

Observation involves watching and recording how something happens in its natural setting, which provides insights that interviews and documents alone cannot always capture (Kumar, 2022). The researcher accompanied CHWs on home visits and observed the full community monitoring workflow over a period of two weeks, from patient visit documentation through facility-level data compilation and supervision meetings. The observations revealed that CHWs carry worn paper registers to every visit, spend over one hour per monthly report manually copying data into facility summary sheets, and have no way of knowing a patient has missed doses until a physical visit occurs. These findings were particularly important for confirming the need for offline mobile functionality, automated missed-dose alerts, and the AI-powered priority list that tells CHWs which patients need attention each day.

Documentation Analysis

Documentation refers to written materials that describe how something works or how it should be used, and reviewing existing documentation is a standard way to build knowledge before designing a new system (Wikipedia, 2024). A review of institutional documents was conducted including patient treatment registers, CHW monthly summary forms, facility supervision checklists, stock record cards, national HIV and TB program reporting templates, and the Rwanda Ministry of Health Digital Health Strategic Plan. This review helped identify existing data structures and field-level data elements, clarified how data flows between community and facility levels, and informed the design of database entities, relationships, and FHIR resource mappings for the proposed system. The Rwanda HIV and TB program guidelines were also reviewed to ensure that system-generated adherence thresholds, clinical alert triggers, and reporting indicators align with national clinical standards.

Expected Results

Based on the objectives set out in this chapter, the following outcomes are expected from the completed system:

- A fully working HIV and TB patient monitoring system in which community-collected patient data is captured on the Flutter mobile application, synchronized

to the cloud backend, processed by the AI engine, and made available to facility clinical staff within minutes of synchronization rather than days or weeks.

- An AI-powered adherence risk dashboard that shows a risk score for every patient reflecting their likelihood of treatment interruption, a priority list telling each CHW which patients need attention today, and trend graphs showing adherence patterns over time.
- A false confirmation detection system that monitors patient medication confirmation patterns, enforces confirmation time windows, and flags discrepancies between patient self-reports and CHW pill counts, with automatic escalation to clinical staff when suspicious patterns are detected.
- An early warning system that detects population-level adherence clusters across villages and CHW networks, giving supervisors advance warning of emerging treatment failure patterns before they become program-level crises.
- Automated medication stock forecasting that predicts when each CHW will exhaust their supply of each medication and generates resupply requests automatically before stockouts occur.
- A web dashboard that facility providers and supervisors without technical backgrounds can use confidently, confirmed through usability testing with real users, displaying real-time adherence metrics, clinical alerts, and CHW performance indicators.
- Documented evidence from testing that all key functions work correctly and that the system meets the performance, security, and reliability standards identified in the requirements, including offline operation, FHIR synchronization, and AI scoring accuracy.

Organization of the Report

Chapter One provides a general introduction to the study, covering the background of Rwanda's HIV and TB community care context, the problem statement, the choice and motivation of the study, objectives, scope, data collection methodology, and expected results.

Chapter Two analyses the current patient monitoring system at Dream Medical Center, identifies its operational limitations across documentation, data quality, care coordination, and stock management, and defines the functional and non-functional requirements that the new system must satisfy.

Chapter Three presents the detailed design of the new system using UML diagrams including use case diagrams, class diagrams, sequence diagrams, and activity diagrams, followed by the AI module specifications, the relational database schema and data dictionary, and the complete system architecture including the Flutter mobile layer, Spring Boot backend, Python AI microservice, and FHIR integration layer.

Chapter Four describes the implementation of the system, covering the technology stack decisions with rationale, screenshots of all key user interfaces for each actor role, and the results of unit testing, integration testing, validation testing, and security testing.

Chapter Five summarizes the project findings, evaluates the degree to which each objective was achieved, reflects on lessons learned during implementation, and provides recommendations for future enhancement of the system.

CHAPTER 2

ANALYSIS OF CURRENT SYSTEM

Introduction

This chapter examines how patient monitoring currently operates at Dream Medical Center, with the aim of understanding what the existing system does well, where it falls short, and what a new system would need to do in order to address those shortcomings. The information in this chapter comes from the interviews, site observations, and document review described in Chapter One. The chapter follows a structured analytical approach: it begins by describing the environment in which the system operates, explains how the current system works, identifies its main limitations, and then translates those findings into clear requirements for the new system.

Description of the Current System Environment

Historical Background

Dream Medical Center is a community-facing health facility located in Kigali, Rwanda, providing integrated HIV and TB care services to patients across its defined catchment area. The CHW network has been central to Dream Medical Center's ability to support treatment adherence among patients who face geographic, economic, or social barriers to regular facility attendance. However, the documentation infrastructure supporting CHW activities has not kept pace with program growth, creating the operational challenges that motivate this project.

Vision

To serve the community and individuals with excellence while proclaiming Christian principles.

Mission

To create a sustainable healthcare model of compassionate excellence, serving patients' physical, emotional, and spiritual needs for the glory of God.

Description of the Current System

The existing patient monitoring process at Dream Medical Center is conducted entirely through manual paper-based mechanisms. When a new HIV or TB patient is enrolled in community care, their demographic and clinical information is recorded by the enrolling CHW in a dedicated paper patient register containing pre-printed fields for name, date of birth, sex, diagnosis, treatment start date, assigned medication, and household location.

During routine home visits, the CHW records the visit date, whether the patient took their prescribed medication, any symptoms reported, side effects experienced, and any psychosocial concerns noted. This information is handwritten in the relevant fields of the paper register at the time of the visit, or in some cases reconstructed from memory shortly afterward. At periodic intervals, typically at monthly supervision meetings, the CHW brings their paper register to the health center where a data clerk manually copies selected information into facility-level summary registers. Clinical staff then review the facility register during planning sessions to identify patients flagged for follow-up. There is no automated forecast, no electronic stock visibility for facility pharmacists, and no alert mechanism for approaching stockouts.

Analysis of the Existing System

Examining the current system against the demands placed on it reveals several important weaknesses. The most fundamental is the complete absence of real-time visibility into patient adherence between CHW visits. Once a CHW leaves the facility with a patient register, no member of clinical staff has any visibility into what is happening in the community until the CHW returns. A patient who begins missing doses the day after a CHW visit may go undetected for the entire following month, by which point the treatment gap may have grown to a clinically significant size.

A second weakness is that the decisions made during home visits are not supported by any data history or analytical tools. CHWs assess patient condition based on what they observe during a single visit, with no way to compare today's situation against previous visits, identify a gradual decline in adherence over time, or receive any alert that a patient's situation has changed since the last visit. While experienced CHWs develop good instincts, these assessments are inevitably less reliable than data-driven analysis and are more likely to miss problems that develop slowly between visits.

Modeling of Current System

The diagram below illustrates the current manual workflow at Dream Medical Center, showing how information flows across the four key actors: the Community Health Worker, the Health Facility, the Supervisor, and the Pharmacy. The fundamental constraint of this system is that data only moves when a person physically carries it. There are no automated alerts, no real-time data exchange, and no digital audit trail at any stage of the process.

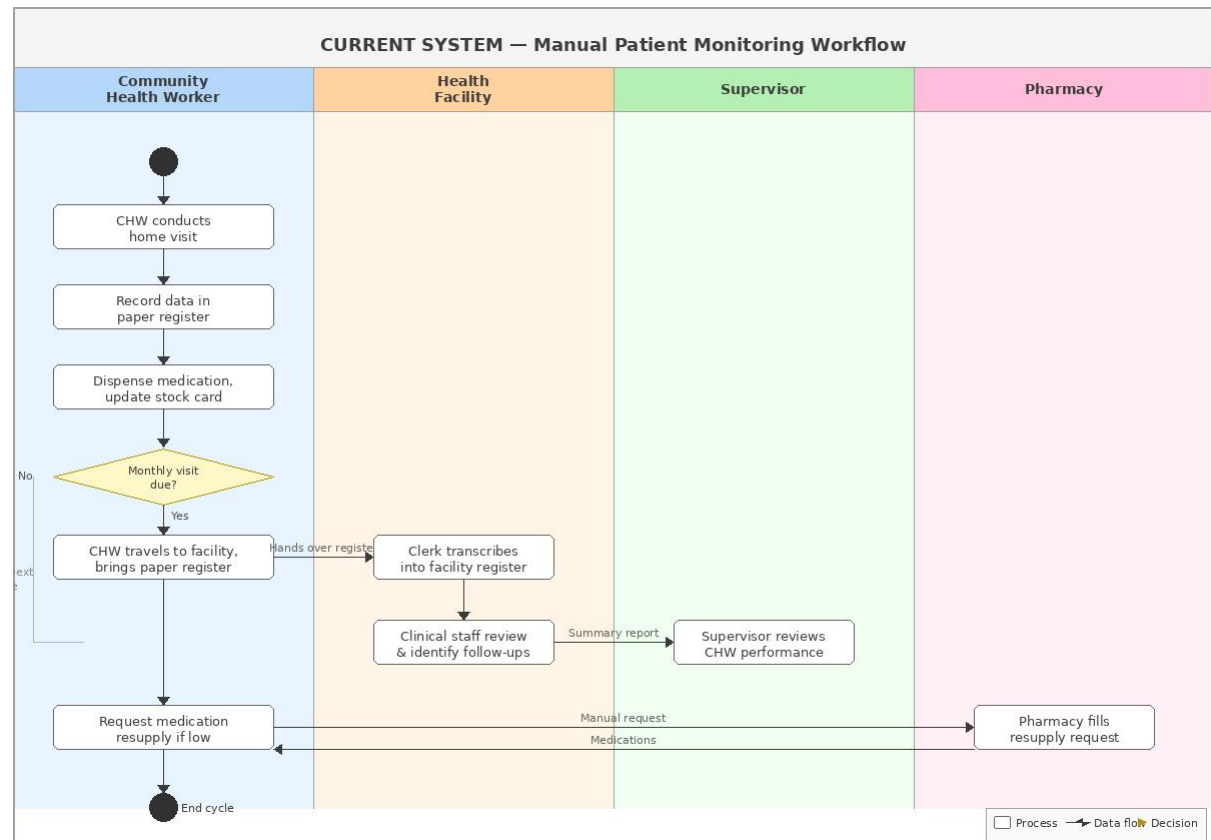


Figure 1: Modeling of Current System

Problems of the Current System

The current patient monitoring approach at Dream Medical Center is largely manual, fragmented, and reactive. CHWs rely entirely on physical visits and paper records to monitor treatment adherence, making it impossible to maintain consistent oversight of patients between visits, especially when adherence problems develop gradually or when a patient's condition changes suddenly between scheduled home visits.

A major problem is the complete lack of real-time data access at the facility level. Patient information is either not collected digitally at all or arrives weeks after the point of care, making it impossible for clinical staff to track adherence trends, identify at-risk patients,

or make informed decisions based on current community conditions. This limitation reduces the ability of facility providers to respond quickly to missed doses, emerging side effects, or deteriorating patient conditions.

In addition, the current system has no predictive or analytical capability whatsoever. Without intelligent decision-support features, CHWs and supervisors cannot detect adherence problems early enough to prevent treatment failure. Manual stock management also leads to preventable medication stockouts that interrupt treatment continuity. As a result, patient outcomes are negatively affected, CHW productivity remains low, and the full potential of Dream Medical Center's community health program is not realized.

Proposed Solutions

The proposed HIV and TB Patient Monitoring System addresses the operational challenges of the current manual system by integrating offline-first mobile technology, cloud-based data processing, artificial intelligence, and FHIR-based health information exchange. Each identified problem is resolved through a specific technical component that ensures accurate monitoring, timely intervention, and intelligent clinical decision support.

- Real-time adherence monitoring: Patient medication confirmations are recorded through the Flutter mobile application immediately at the point of care and synchronized to the cloud backend automatically when connectivity is restored, giving clinical staff up-to-date adherence information without waiting for monthly CHW visits.
- AI-powered risk prediction: A Python-based AI microservice runs an adherence risk scoring model every night that evaluates each patient's confirmation history, response time patterns, side effect reports, and clinical outcomes to calculate a risk score and predict which patients are most likely to interrupt treatment in the next 7 to 14 days.
- Smart CHW prioritization: Every morning the system generates an AI-powered priority list for each CHW, ranking their patients by risk level and telling them exactly which patients need a visit today, which need a phone call, and which are stable. This allows CHWs to focus their limited time on the patients who need them most.

- False confirmation detection: The system monitors patient medication confirmation patterns using timestamp analysis, confirmation window enforcement, and pill count reconciliation during CHW visits. When suspicious patterns are detected the system automatically escalates to clinical staff for review.
- Automated stock forecasting: The AI engine calculates a days-remaining estimate for each medication held by each CHW based on current stock levels and dispensing rates, and automatically generates resupply requests before stockouts occur.
- Early warning for treatment failure: The system monitors adherence patterns across the entire patient population and flags geographic or CHW-level clusters of declining adherence, giving supervisors advance warning of emerging program-level problems before they become crises.
- FHIR-based data exchange: All patient and clinical data is exchanged with hospital Electronic Health Record systems using the HL7 FHIR R4 standard through a HAPI FHIR gateway, enabling clinical outcome correlation and ensuring that community-collected data is immediately accessible to facility providers.
- Offline operational reliability: The Flutter mobile application stores all patient data locally using SQLite and operates fully without internet connectivity. Data synchronizes automatically with the server when connectivity is restored, without requiring any action from the CHW.

System Requirements

The HIV and TB Patient Monitoring System is designed to support community-level patient monitoring, store and analyze adherence data, and enable intelligent early intervention to prevent treatment failure. By integrating offline mobile technology, cloud-based AI processing, FHIR-based data exchange, and role-specific dashboards, the system enables CHWs, facility providers, and supervisors to monitor patient care in real time while reducing manual workload. The following functional requirements define the core capabilities the system must provide.

Functional Requirements

- **REQ-01:** The System Administrator shall be able to register and manage user accounts for all roles within the system.
- **REQ-02:** The system shall allow registered users (Administrator, Supervisor, Facility Provider, CHW, and Patient) to securely log in and access role-specific functionalities.
- **REQ-03:** The CHW shall be able to register new HIV and TB patients offline using a structured mobile form capturing full demographics, diagnosis details, and treatment parameters.
- **REQ-04:** The CHW shall be able to record home visit outcomes including adherence status, symptom observations, side effect reports, and psychosocial notes.
- **REQ-05:** The system shall calculate and display rolling adherence percentages automatically without requiring any manual calculation by the CHW.
- **REQ-06:** The system shall send medication reminder notifications to patients at their scheduled dose times via Firebase Cloud Messaging or SMS through Africa's Talking.
- **REQ-07:** The system shall allow patients to confirm medication taken through a simple mobile interface within a defined confirmation time window.
- **REQ-08:** The AI engine shall run an adherence risk scoring model every night and assign each patient a risk level of Low, Moderate, High, or Critical based on their confirmation history, response time patterns, side effect reports, and clinical outcomes.
- **REQ-09:** The system shall generate an AI-powered daily priority list for each CHW ranking their patients by risk level and specifying the recommended action for each patient.
- **REQ-10:** The system shall monitor patient medication confirmation patterns and flag suspicious behavior including timestamp anomalies, out-of-window confirmations, and pill count discrepancies.
- **REQ-11:** The system shall automatically escalate false confirmation alerts to facility providers when two or more suspicious signals are detected within a 14-day period.

- **REQ-12:** The system shall give facility providers access to a real-time patient dashboard displaying current adherence trends, recent visit histories, AI risk scores, and active clinical alerts.
- **REQ-13:** The system shall correlate patient self-reported adherence with facility lab results received through FHIR and generate a clinical-behavioral discrepancy alert when high reported adherence is inconsistent with worsening clinical outcomes.
- **REQ-14:** The system shall track CHW medication stock levels and calculate a days-remaining estimate for each medication based on current stock and dispensing rates.
- **REQ-15:** The system shall automatically generate a resupply request and notify the facility pharmacy when a CHW's medication stock falls below a configured threshold.
- **REQ-16:** The AI engine shall monitor adherence patterns across the entire patient population and generate an early warning alert when geographic or CHW-level clusters of declining adherence are detected.
- **REQ-17:** The system shall provide supervisors with aggregate performance dashboards showing CHW activity levels, patient load distributions, adherence averages, and stock availability summaries.
- **REQ-18:** The system shall automatically synchronize all locally stored data with the server within 60 seconds of network detection without requiring any user action.
- **REQ-19:** The System Administrator shall be able to review a comprehensive audit log recording all data creation, modification, and access events.

Non-Functional Requirements

- **REQ-20 (Availability):** The mobile application must support full core functionality without internet connectivity. All patient registration, visit recording, adherence tracking, and stock management features must operate offline using local SQLite storage.
- **REQ-21 (Reliability):** Data synchronization must initiate automatically within 60 seconds of network detection without requiring any user action. All records created or modified offline must be synchronized completely.

- **REQ-22 (Security):** All data must be encrypted using AES-256 at rest and TLS 1.3 in transit. No patient data may be stored in plain text on any device. Failed authentication attempts must be logged and rate-limited.
- **REQ-23 (Security):** Each user role must be able to access only the data and functions appropriate to their defined responsibilities. Cross-role data access must be prevented at the API layer.
- **REQ-24 (Usability):** The mobile interface must be operable by a CHW with basic smartphone literacy after no more than two hours of initial training. Key actions must require no more than three taps from the home screen. Kinyarwanda language support is required.
- **REQ-25 (Interoperability):** All external data exchange must use HL7 FHIR R4 resources. The system must pass FHIR R4 resource validation for Patient, Observation, MedicationStatement, and CarePlan resource types.
- **REQ-26 (Performance):** Patient record retrieval must complete within 3 seconds on supported mobile devices. Server API responses must be returned within 2 seconds under normal load conditions.
- **REQ-27 (Performance):** The full cycle from patient confirmation to AI risk score update must complete within 60 seconds of server synchronization.
- **REQ-28 (Scalability):** The system architecture must support a minimum of 500 concurrent CHW mobile users and 50 concurrent web dashboard users without architectural redesign.
- **REQ-29 (Intelligence):** The adherence risk scoring model must achieve a minimum precision of 80% in identifying high-risk patients, validated against CHW visit outcomes during the testing phase.
- **REQ-30 (Legal):** The system must comply with Rwanda Law No. 058/2021 of 13/10/2021 on the protection of personal data and privacy, including user consent management and data subject rights.

CHAPTER 3

REQUIREMENTS ANALYSIS AND DESIGN OF THE NEW SYSTEM

Introduction

This chapter presents the comprehensive requirements analysis and system design of the HIV & TB Patient Monitoring System. The design phase translates the requirements identified in Chapter Two into a set of models and diagrams that describe how the system will work, how its parts relate to each other, and how data will flow through it. The Unified Modeling Language is used throughout this chapter as the standard notation for system design, following the Object-Oriented Methodology (Larman, 2009). UML provides a clear, internationally recognized way of describing software systems through diagrams, making it easier for developers, supervisors, and stakeholders to understand and evaluate the proposed design before any code is written.

Unified Modeling Language

The Unified Modeling Language is a standardized visual language for describing software systems. It uses a set of diagram types, each designed to show a different aspect of a system. A use case diagram shows who uses the system and what they can do with it. A class diagram shows the main things the system keeps track of and how they relate to each other. A sequence diagram shows the order in which different parts of the system communicate when carrying out a specific task. An activity diagram shows the flow of steps in a process, including where decisions are made. A database diagram shows how information is organized and stored. Together these diagrams give a complete picture of a system at a level of detail that is precise enough to guide implementation but clear enough to be reviewed and understood by non-technical stakeholders (Larman, 2009).

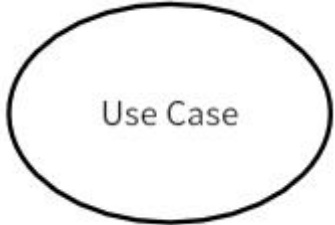
By using OOM and UML to design the new system, this project ensures that the design is systematic, well-documented, and easy to extend or modify in the future. The following sections present each diagram type in turn, with an explanation of what each diagram shows and why the design decisions it reflects were made.

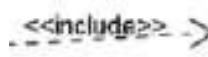
Design of the New System Diagrams

Use Case Diagram

A use case diagram shows the system from the outside looking in who uses it and what they can do with it. Each oval shape represents a function of the system, called a use case. Each stick figure represents a type of user, called an actor. The lines connecting actors to use cases show which users can access which functions. The rectangle drawn around the use cases show which users can access which functions. The rectangle drawn around the use cases represents the boundary of the system itself, separating the functions that the system provides from the external actors who use it (Larman, 2009).

The HIV & TB Intelligent Patient Monitoring and Early Intervention System has six actors. The Patient is the primary beneficiary of care who receives medication reminders, confirms doses, and views their own treatment summary. The Community Health Worker is the primary field user who registers patients, records home visits, manages medication stock, and acts on AI-generated priority lists. The Facility Provider is the clinical user who monitors patient dashboards, receives AI-generated alerts, and manages referrals. The Supervisor oversees the CHW network and receives population-level early warning alerts. The System Administrator manages user accounts, configures system parameters, and monitors system health. The System itself acts as an automated actor responsible for AI processing, alert generation, FHIR synchronization, and notification dispatch. All human actors share the login function, which is the entry point to the system for every user.

Description	Phase
Use Case ✓ A use case represents a specific function or operation that the system performs, either automatically by the system itself or manually by a human actor. ✓ Naming convention: Verb + Noun (e.g., Register Patient, Generate Alert, Confirm Medication).	

✓ Every actor in the system must be connected to at least one use case.	
Boundary of the system ✓ The system boundary is represented by a rectangle that encloses all use cases belonging to the system. ✓ It separates the internal functions of the system from the external actors who interact with it. ✓ The boundary defines the scope of the system as specified in the requirements.	
Association Link ✓ An association in a use case diagram represents the interaction between an actor and a use case, showing the actor's involvement in that function. ✓ Actors and use cases are connected solely through association links. ✓ Association relationships are also referred to as communication associations.	
An Include Relationship ✓ Denotes the incorporation of a use case's capabilities into another. ✓ Features a drawn arrow connecting the used use case to the base use case.	

Use Case Diagram

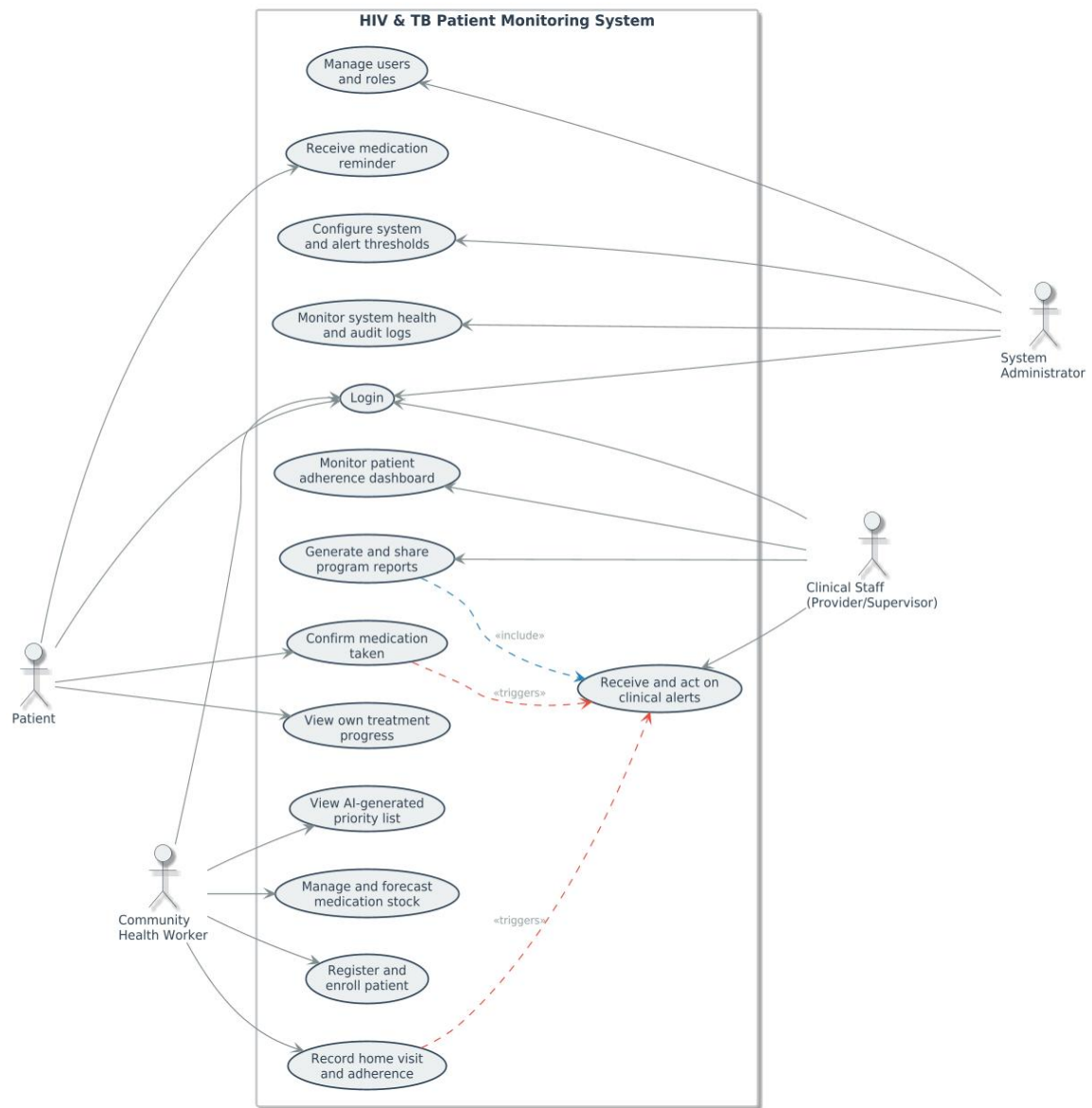


Figure 2: Use case Diagram

Use Case Descriptions

Use case descriptions provide more detail about what happens when a use case is carried out. Each description follows a standard format that lists the name, the actors involved, the purpose, the condition that must be true before the use case can begin, the condition that is true after it completes, the normal steps, and any alternative steps for when something goes wrong.

Field	Details
Name	Login
Actors	Patient, Community Health Worker, Clinical Staff, System Administrator
Description	Allows a registered user to sign in and access functions specific to their role.
Pre-condition	The user has an active account created by the System Administrator.
Post-condition	The user is authenticated and redirected to their role-specific dashboard.
Normal flow	1. User opens the app or web portal and enters email and password. 2. System validates credentials via HTTPS. 3. System issues a JWT token and loads the role-specific dashboard.
Alternative Flow	If credentials are incorrect, the system shows an error and denies access. After three failed attempts the account is locked.

Table 1: Use Case Description for Login

Field	Details
Name	Receive Medication Reminder and Confirm Medication Taken
Actors	Patient
Description	System sends a scheduled reminder to the patient. Patient confirms the dose was taken within the allowed time window.
Pre-condition	Patient has an active treatment plan with scheduled dose times.
Post-condition	Confirmation is recorded with timestamp. Adherence record is updated and CHW is notified.
Normal flow	1. System sends a push notification or SMS at the scheduled dose time. 2. Patient opens the app and taps confirm within 45 minutes. 3. System records the confirmation and updates the adherence record.
Alternative Flow	If the patient does not confirm within 45 minutes, the system

	records a missed dose and alerts the CHW.
--	---

Table 2: Use Case Description for Receive Medication Reminder

Field	Details
Name	View Own Treatment Progress
Actors	Patient
Description	Allows the patient to view their adherence history, current treatment stage, and next scheduled dose.
Pre-condition	Patient is logged into the mobile app and has an active treatment plan.
Post-condition	Patient views their adherence percentage, missed doses, and next dose time.
Normal flow	1. Patient navigates to My Progress. 2. System displays adherence percentage, dose calendar, and next scheduled dose.
Alternative Flow	If the device is offline, the system displays the last synchronized data with a timestamp.

Table 3: Use Case Description for View Own Treatment Progress

Field	Details
Name	Register and Enroll Patient
Actors	Community Health Worker
Description	Allows the CHW to register a new HIV or TB patient using a structured offline-capable mobile form.
Pre-condition	CHW is logged into the mobile app. Device may be offline.
Post-condition	Patient record is saved locally, a unique ID is assigned, and the record is queued for FHIR sync.
Normal flow	1. CHW selects Register New Patient. 2. CHW enters demographics, diagnosis type, and treatment start date. 3. System checks for duplicates, assigns a patient ID, and saves locally.
Alternative Flow	If a duplicate is detected, the system displays the matching record

	and prompts the CHW to confirm or link.
--	---

Table 4: Use Case Description for Register and Enroll Patient

Field	Details
Name	Record Home Visit and Adherence
Actors	Community Health Worker
Description	Allows the CHW to document visit outcomes including pill count, adherence status, side effects, and next visit schedule.
Pre-condition	The patient exists in the system and is assigned to the CHW.
Post-condition	Visit record is saved. Adherence score is recalculated. AI priority list is refreshed.
Normal flow	1. CHW selects the patient and taps Record Home Visit. 2. CHW enters pill count, adherence status, symptoms, and notes. 3. System saves the record, recalculates adherence, and refreshes the priority list.
Alternative Flow	If pill count does not match confirmed doses, a discrepancy alert is generated. If a serious adverse event is reported, the system prompts an urgent referral.

Table 5: Use Case Description for Record Home Visit and Adherence

Field	Details
Name	View AI-Generated Priority List and Manage Medication Stock
Actors	Community Health Worker
Description	CHW views a daily AI-ranked patient list and manages medication stock with automated resupply forecasting.
Pre-condition	CHW is logged into the app. AI nightly scoring run is complete. Stock records exist.
Post-condition	CHW knows which patients to prioritize. Stock is updated. Resupply request is auto-submitted if stock falls below 14 days.
Normal flow	1. CHW opens the app and views the priority list grouped into Visit Today, Call Today, and Stable. 2. CHW navigates to Stock, reviews levels and days-remaining estimates.

	3. CHW records a dispensing event. System updates stock and forecasts.
Alternative Flow	If AI scoring has not run, the previous day's list is shown with a warning. If stock is zero, dispensing is blocked and the resupply request is escalated as urgent.

Table 6: Use Case Description for View AI-Generated Priority List

Field	Details
Name	Monitor Patient Adherence Dashboard and Receive Clinical Alerts
Actors	Clinical Staff
Description	Clinical Staff view a real-time dashboard of all community patients and receive AI-generated alerts when clinical lab results contradict reported adherence.
Pre-condition	Clinical Staff is logged into the web portal. Patients are assigned to the facility.
Post-condition	Clinical Staff has current visibility into all patients. Alerts are acknowledged and referrals initiated where needed.
Normal flow	1. Staff opens the Patient Dashboard and reviews adherence trends and risk levels. 2. When a lab result arrives via FHIR, the AI compares it against adherence records. 3. If a discrepancy is detected, an alert is sent via push notification and email. 4. Staff acknowledges the alert and initiates a clinical review or referral.
Alternative Flow	If an alert is not acknowledged within 48 hours, the system re-triggers it and escalates to supervisor level.

Table 7: Use Case Description for Monitor Patient Adherence Dashboard

Field	Details
Name	Generate and Share Program Reports
Actors	Clinical Staff
Description	Allows Clinical Staff to generate program performance reports and share them automatically via email.
Pre-condition	Clinical Staff is logged into the web portal. Data exists for the selected period.

Post-condition	Report is generated in PDF or Excel format and sent to the registered email address.
Normal flow	1. Staff selects report type and date range. 2. System aggregates adherence, CHW activity, and stock data. 3. Staff exports the report. System sends it automatically to the registered email.
Alternative Flow	If some CHW data is missing, the system flags incomplete records and includes a note in the report.

Table 8: Use Case Description for Generate and Share Program Reports

Field	Details
Name	Manage Users, Configure System and Monitor Health
Actors	System Administrator
Description	Administrator manages all user accounts, configures system settings and alert thresholds, and monitors system health and audit logs.
Pre-condition	System Administrator is logged into the administration portal.
Post-condition	Accounts are created with encrypted credentials. Settings are applied. System health and audit trail are visible.
Normal flow	1. Admin creates a user account, assigns a role, and system sends a welcome email. 2. Admin configures alert thresholds, medication lists, and FHIR endpoint settings. 3. Admin reviews system health indicators and audit logs for all data events.
Alternative Flow	Deactivating a user immediately revokes all access and is logged in the audit trail. If a system component goes offline, the Administrator is alerted automatically.

Table 9: Use Case Description for System Administrator Functions

Class Diagram

A class diagram shows the main concepts that the system needs to keep track of called classes and the relationships between them. Each class is shown as a rectangle divided into three sections: the name at the top, the information it holds in the middle, and the actions it can perform at the bottom. The lines connecting classes show how they relate to

each other, with labels and numbers indicating the nature and multiplicity of each relationship (Bruegge & Dutoit, 2010).

The class diagram for the HIV & TB Patient Monitoring System shows twelve classes. User holds account information and determines what role a person has. CHW, Clinical Staff, and SystemAdmin all inherit from User. Patient holds the core demographic and clinical record for each enrolled patient. Treatment Plan holds the medication regimen and schedule for each patient. HomeVisit records every CHW home visit including pill count and adherence observations. MedicationRecord tracks doses scheduled versus doses taken and stores the computed adherence percentage. Alert holds all system-generated notifications. StockRecord tracks each CHW's medication inventory with AI-generated resupply forecasting. FHIRSync packages local records as FHIR R4 resources and manages data exchange with the facility Electronic Health Record system. AIEngine runs all six intelligent processing modules covering risk scoring, false confirmation detection, priority list generation, clinical outcome correlation, stock forecasting, and cluster detection.

Class Diagram

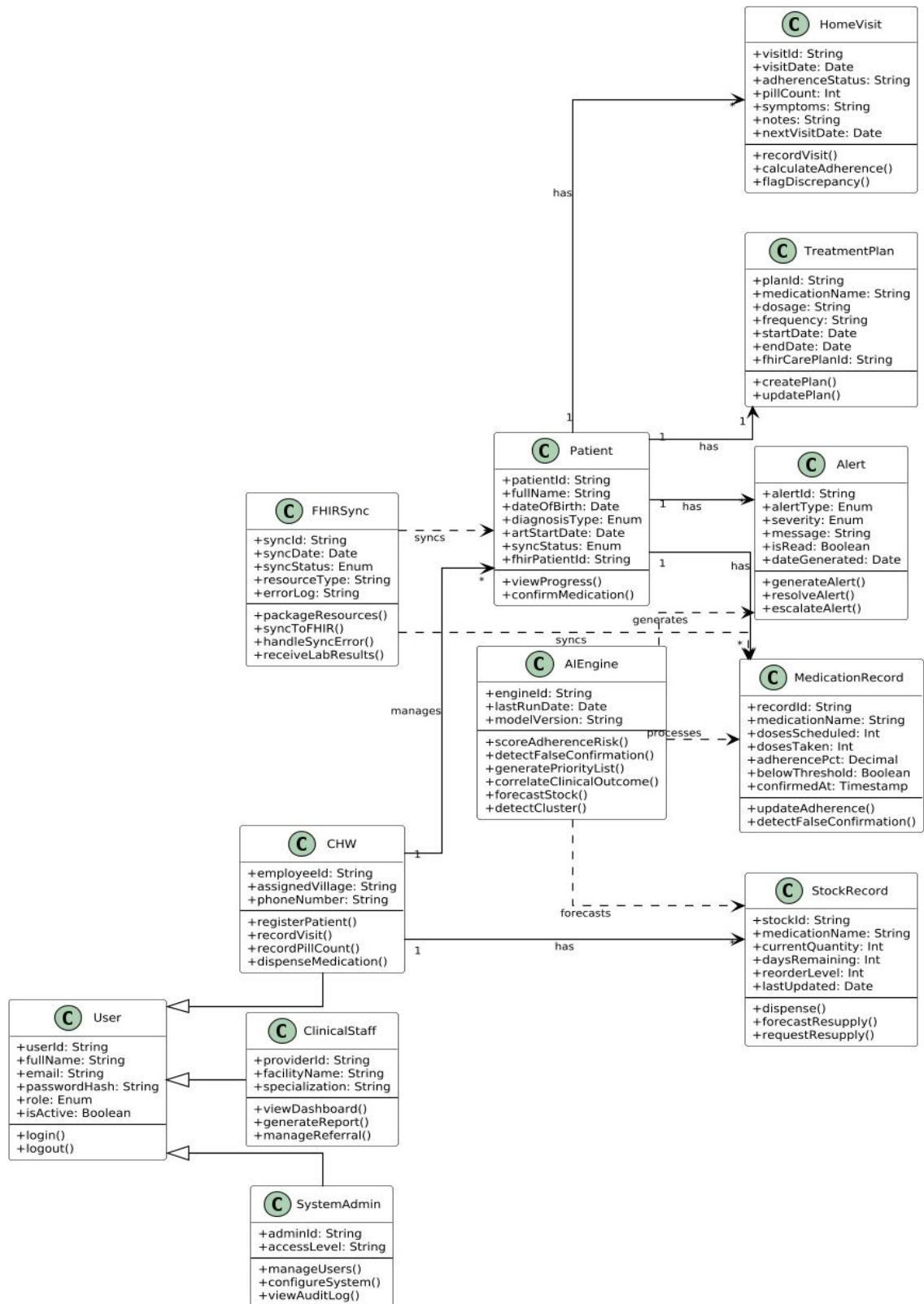
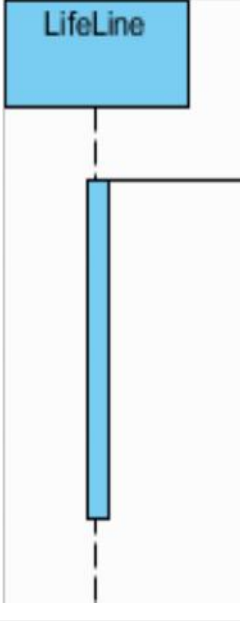
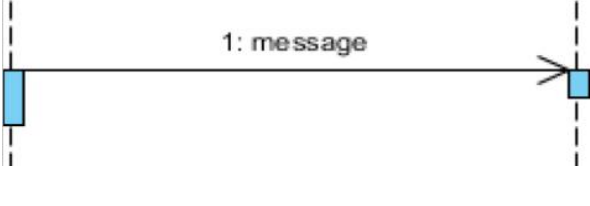
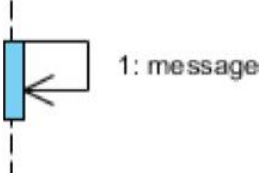
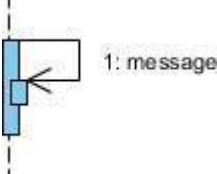
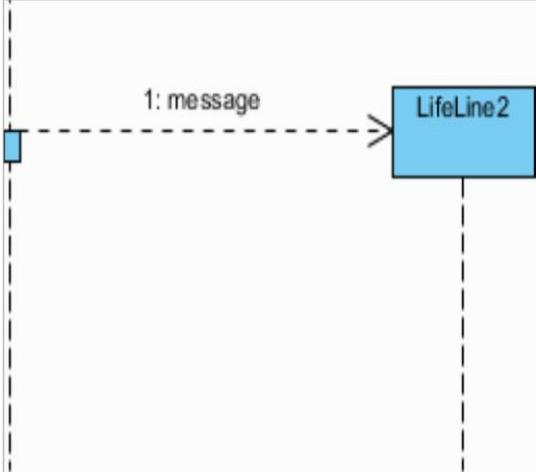


Figure 3:Class Diagram

Sequence Diagrams

A sequence diagram shows the order in which different parts of the system communicate when carrying out a specific task. Each participant in the interaction is shown as a labelled box at the top of a vertical line. Messages between participants are shown as horizontal arrows, numbered in the order they occur. A solid arrow represents a request or command being sent. A dashed arrow represents a response being returned. A small rectangle on a vertical line shows when that participant is actively doing something (Bruegge & Dutoit, 2010).

Description of a notation	Graphic representation
A Lifeline A lifeline in a sequence diagram represents an individual participant or entity involved in the interaction.	
Activations A thin rectangle on a lifeline represents the duration during which an element performs an operation or executes a specific task. The start time is placed at the top of the rectangle and the completion time at the bottom.	
Call Message A message in a sequence diagram represents a specific type of interaction between lifelines, indicating the flow of	

information between them. A call message represents the initiation or activation of a target lifeline to perform a specific action or operation.	
Return Message Return messages represent the response or result from a previous message to the caller, indicating the completion of an operation or the data returned by the invoked method.	
Self-Message Self-messages represent method invocations or interactions within the same lifeline, indicating actions performed by the same object or entity.	
Recursive Message Recursive messages represent the calling of messages from the same lifeline, where the message target is an activation located above the current activation.	
Create Message The create message represents the instantiation or creation of a new lifeline or object during the interaction.	

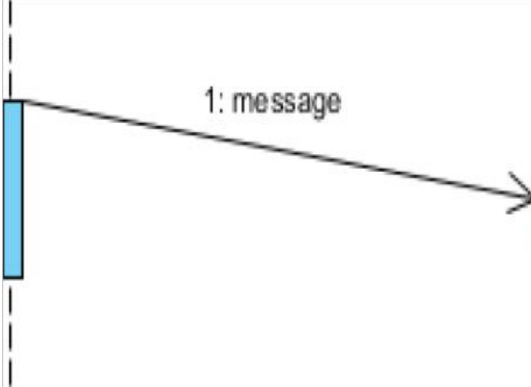
Destroy Message A destruct message indicates the request to terminate or destroy the lifecycle of the target lifeline.	 The diagram shows two vertical dashed lines representing lifelines. The left lifeline has a blue rectangular activation bar. A horizontal arrow points from this bar to the right lifeline, labeled "1: message". The arrow ends in a solid arrowhead pointing to a small blue rectangle on the right lifeline. Below this rectangle, the right lifeline ends with a large 'X' symbol, indicating the destruction of the lifeline.
Duration Message A duration message represents the interval between two specific points when executing a message invocation.	 The diagram shows two vertical dashed lines representing lifelines. The left lifeline has a blue rectangular activation bar. A diagonal arrow points from this bar to the right lifeline, labeled "1: message". The arrow ends in a solid arrowhead pointing to a small blue rectangle on the right lifeline. The right lifeline continues vertically beyond this point.
Note A note in a sequence diagram allows the modeler to add additional information or annotations. These do not directly impact the diagram semantics but provide relevant context or explanations.	 The diagram shows a light blue rectangular box with a black border. The top-right corner of the box is folded over, creating a tab-like shape, which is the standard notation for a note in a sequence diagram.

Table 10:Sequence diagram notation reference

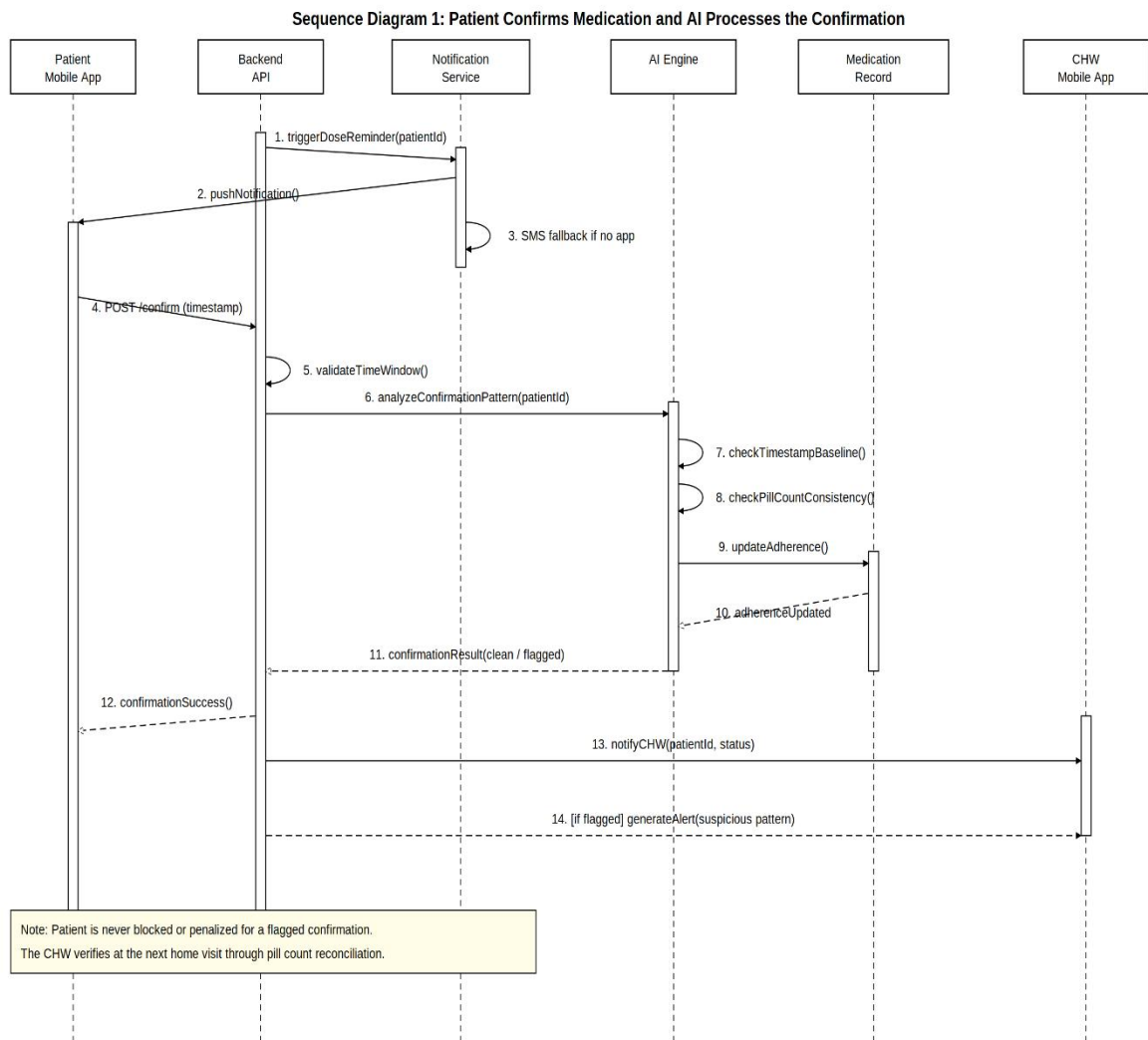


Figure 4: Sequence Diagram 1 Patient Confirms Medication

The sequence begins when the Backend API dispatches a medication reminder to the Patient's device via push notification or SMS. The Patient confirms the dose through the mobile app and the confirmation is sent with a timestamp to the Backend API. The API forwards the event to the AI Engine which evaluates the confirmation pattern against the patient's baseline behavior and pill count history. If the pattern is clean the MedicationRecord is updated and the CHW is notified. If a suspicious pattern is detected the AI Engine flags the record, raises the patient's risk score, and generates an alert to the CHW dashboard.

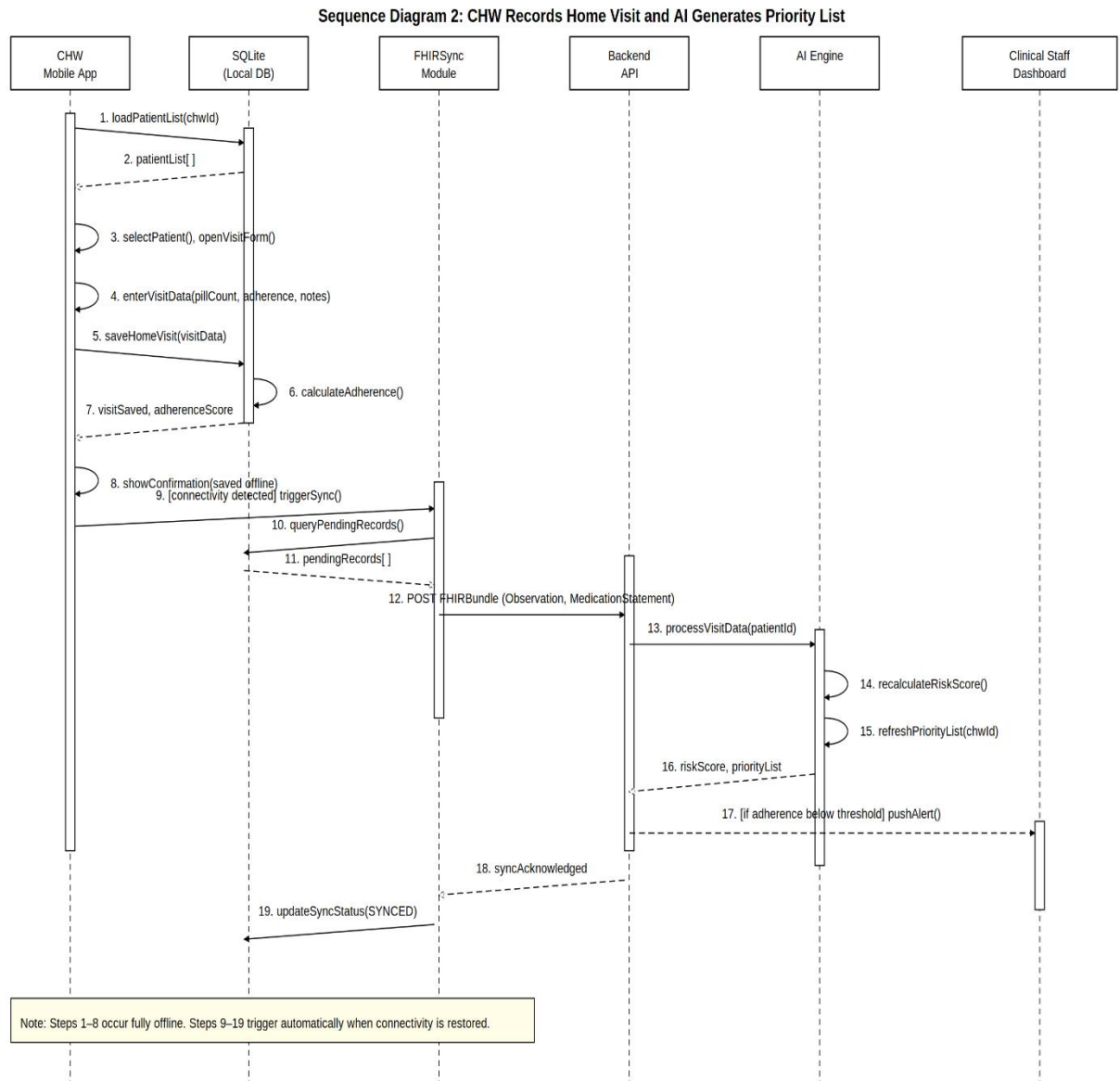


Figure 5: Sequence Diagram 2 CHW Records Home Visit

The sequence begins when the CHW selects a patient from the mobile app and enters visit data including pill count, adherence observations, and symptoms. The app saves the HomeVisit record to the local SQLite database and calculates adherence offline. When connectivity is detected the FHIRSync module automatically packages the pending records as FHIR Observation and MedicationStatement resources and sends them to the Backend API. The API passes the data to the AI Engine which recalculates the patient's risk score and refreshes the CHW priority list. If adherence has dropped below the configured threshold the system pushes an alert to the Clinical Staff dashboard.

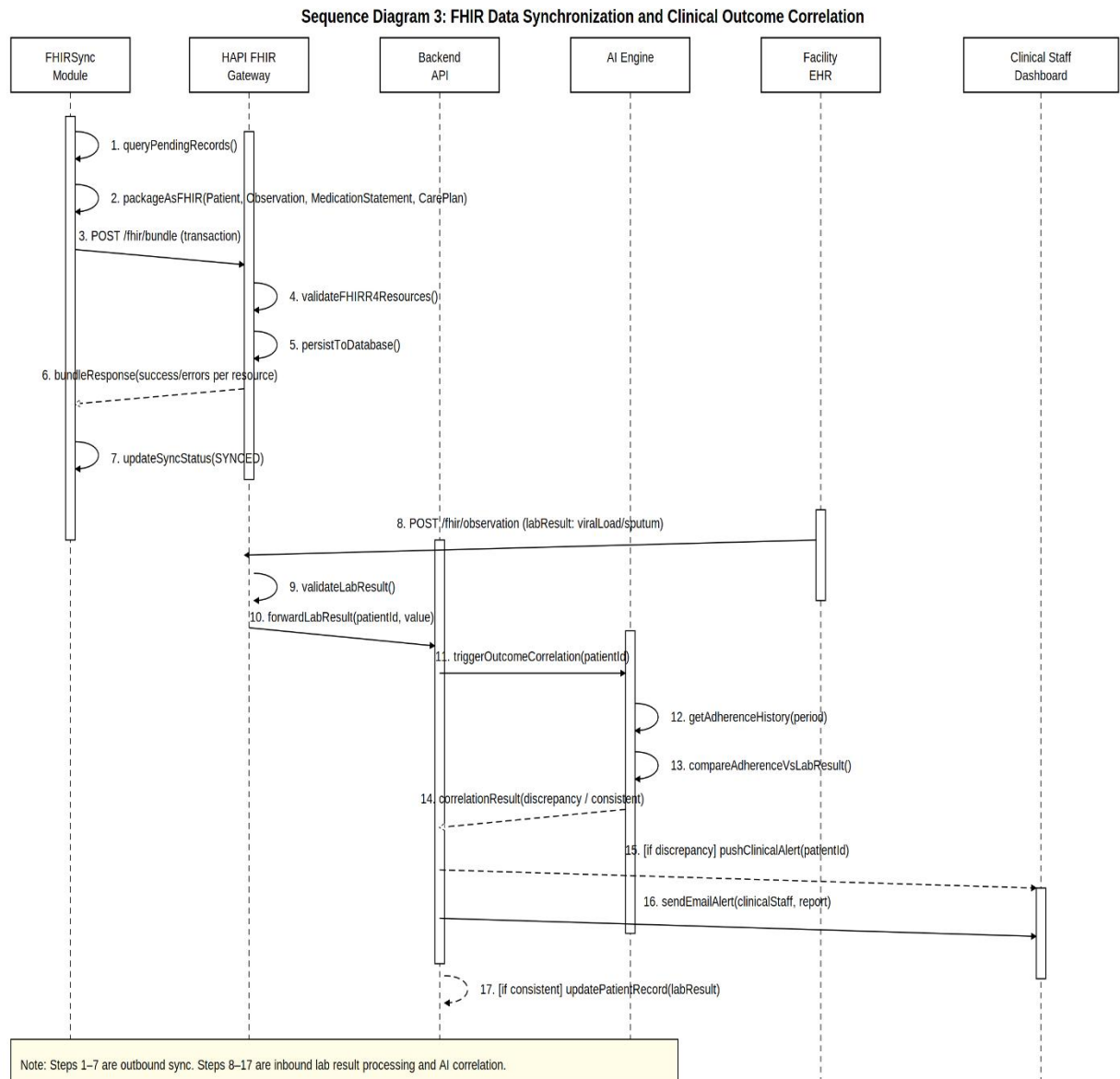


Figure 6: Sequence Diagram 3 FHIR Data Synchronization

The sequence begins when the FHIRSync module detects connectivity and packages all pending local records into a FHIR transaction bundle containing Patient, Observation, MedicationStatement, and CarePlan resources. The bundle is sent to the HAPI FHIR Gateway which validates each resource and stores them in the central database. When the facility EHR pushes a new lab result for an enrolled patient, the Backend API triggers the AI Engine's clinical outcome correlation module. The AI Engine compares the patient's reported adherence against the direction of the new lab result. If a discrepancy is detected the system generates a high-priority clinical alert sent to the Clinical Staff dashboard via WebSocket and email.

Activity Diagrams

An activity diagram shows the flow of steps in a process, similar to a flowchart. It is particularly useful for showing processes that involve decisions, because it makes the different paths through a process easy to see and follow. A filled circle marks the start of the process. Rounded rectangles represent individual steps or actions (Bruegge & Dutoit, UML modeling, design patterns, iterative software processes, 2010).

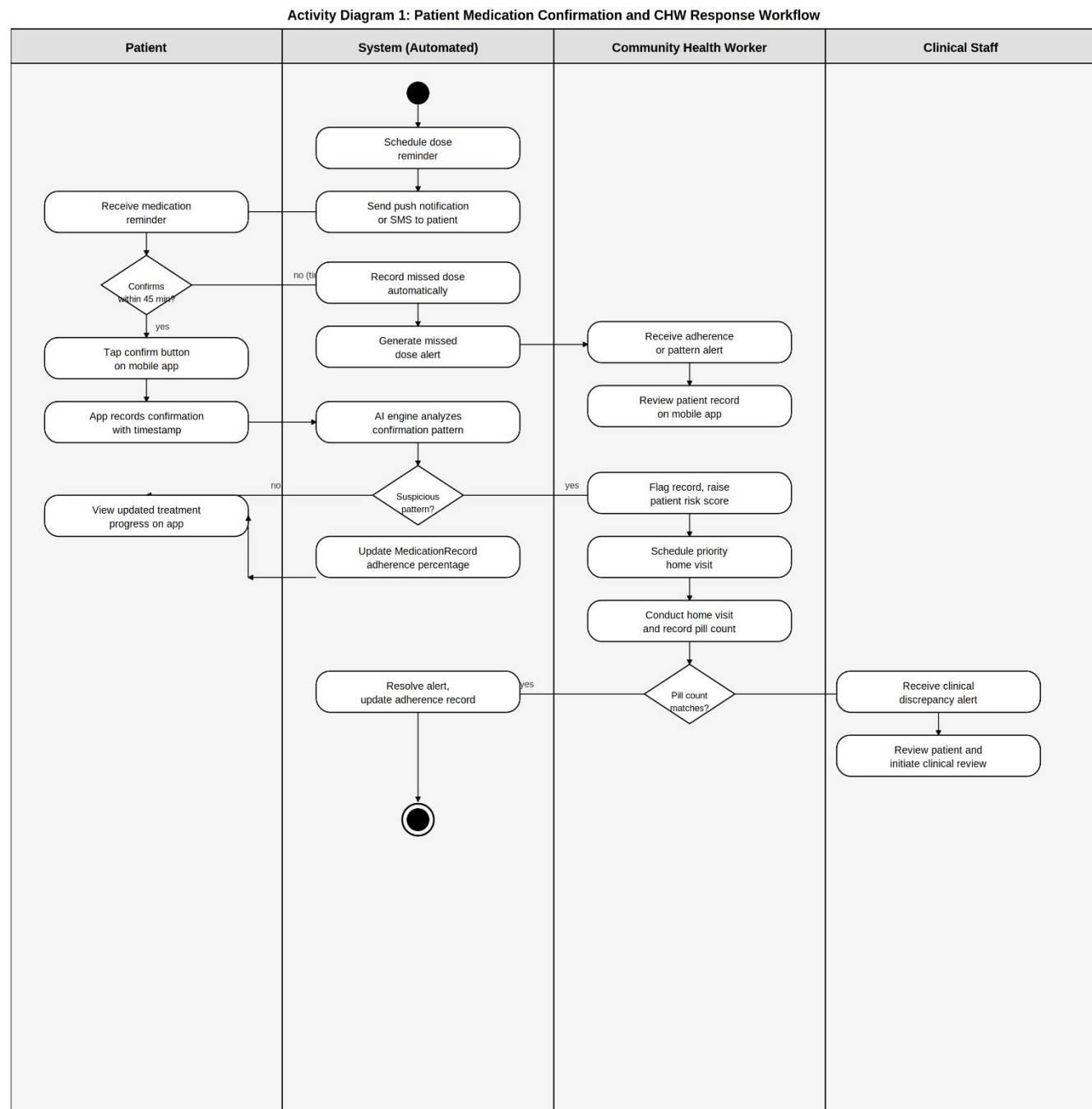


Figure 7: Patient Medication Confirmation and CHW response workflow

Activity Diagram 2: CHW Home Visit, FHIR Sync, and AI Processing Workflow

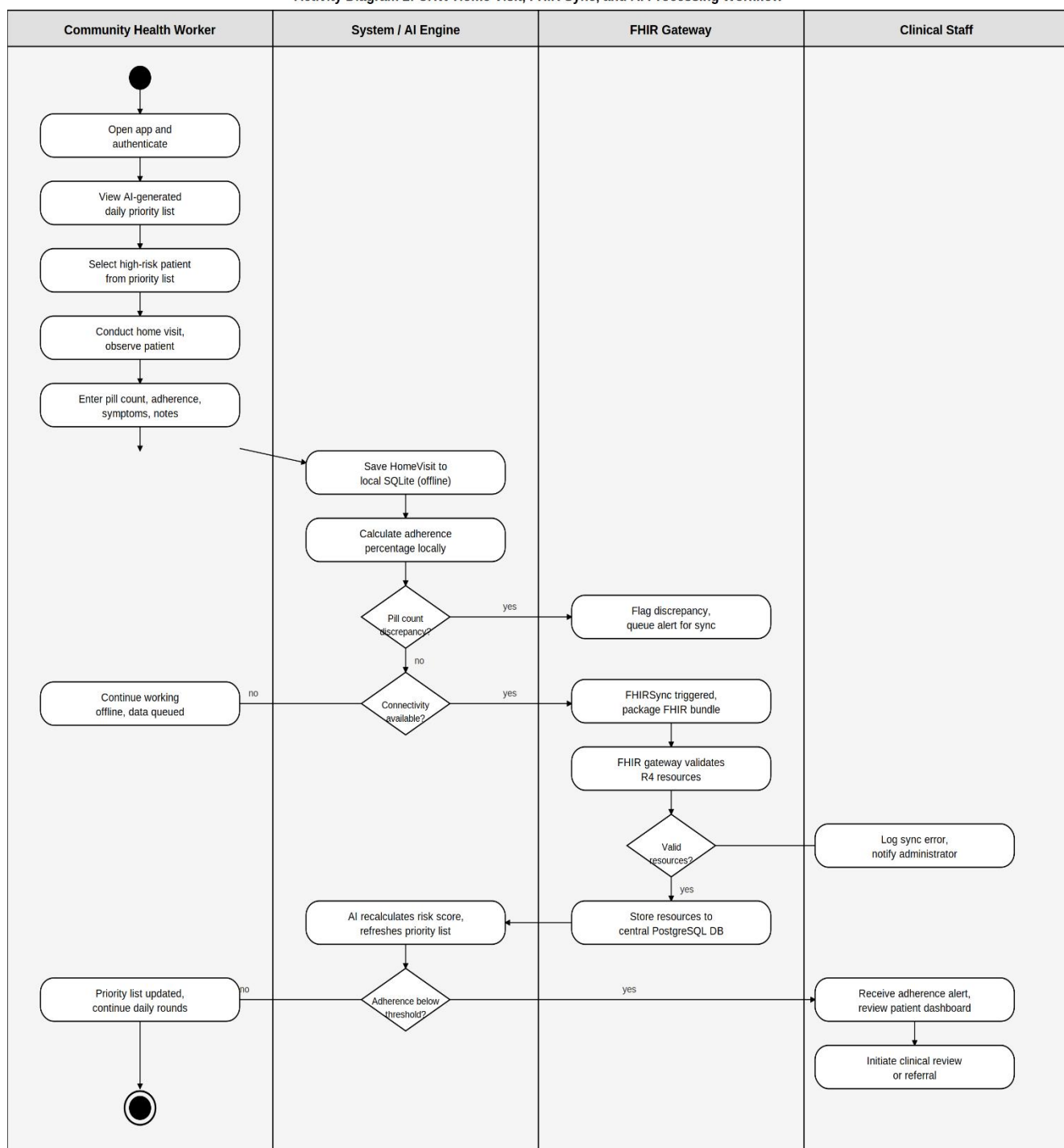


Figure 8:CHW Home visit, FHIR Sync and AI processing Workflow

Database Design

Database System Concepts is a foundational computer science textbook authored by Abraham Silberschatz, Henry F. Korth, and S. Sudarshan. Published by McGraw-Hill Education, it serves as a standard introduction to database systems, emphasizing both theoretical principles and practical implementations. The book is widely adopted in undergraduate and graduate courses worldwide (Silberschatz, Korth, & Sudarshan, 2019). The database is the part of the system that stores all the information it collects and produces. A well-designed database organizes information clearly, avoids storing the same piece of information in more than one place, and makes it easy to retrieve what is needed quickly. The HIV & TB Patient Monitoring System uses a dual-database strategy. A lightweight SQLite database runs on each CHW and patient mobile device, storing only the data relevant to that user so the system works fully offline in areas without connectivity. A central PostgreSQL database runs on the cloud server and holds all synchronized data from every device. Every record created on a mobile device carries a sync status of PENDING until it is successfully transmitted to the server, at which point it is marked SYNCED.

The schema diagram shows the twelve tables in the system and the relationships between them. Each table represents one type of information. For example, the `system_users` table holds authentication details for every user regardless of role, and the `ai_risk_scores` table holds the daily AI-generated risk score and priority ranking for every patient. The arrows between tables show where one table references information held in another, and the multiplicity labels show how many records in one table can relate to records in another.

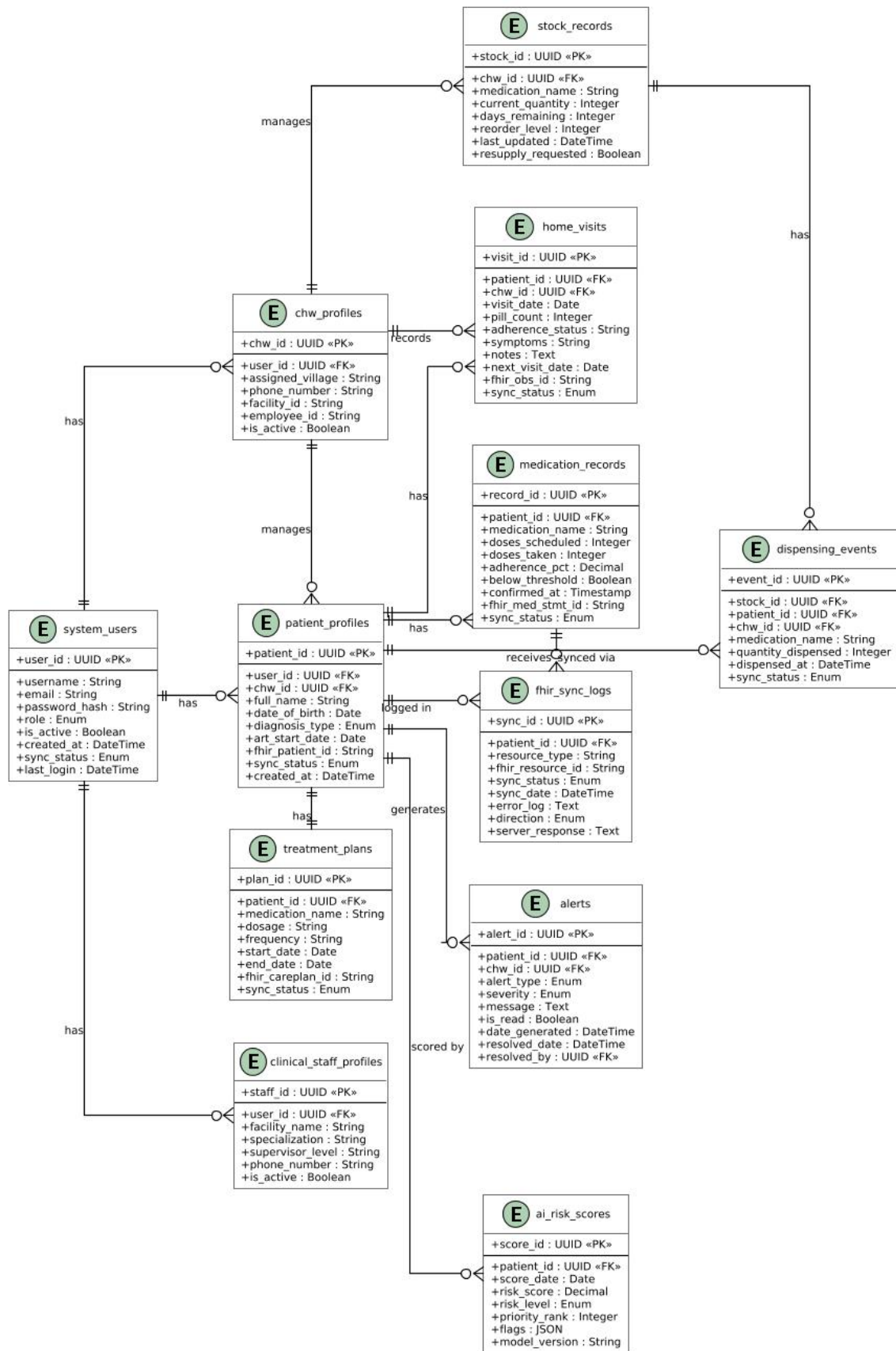


Figure 9:Database Schema diagram

Data Dictionary

A data dictionary is a reference document that describes the purpose, data type, constraints, and meaning of every field in the database (Silberschatz, Korth & Sudarshan, 2019). The following tables describe each table in the HIV & TB Patient Monitoring System database.

Data dictionary for system-users Table

This table is the central authentication table for all system users regardless of role.

Field	Data Type	Constraints	Description
user_id	UUID	Primary Key	Unique identifier for every system user
email	VARCHAR(100)	Unique, NOT NULL	Email address used for login and notifications
password_hash	VARCHAR(255)	NOT NULL	Bcrypt-hashed password — never stored as plain text
role	ENUM	NOT NULL	User role: PATIENT, CHW, CLINICAL_STAFF, ADMIN
is_active	BOOLEAN	Default: true	Whether the account is currently active
created_at	TIMESTAMP	Auto-set	Date and time the account was created
last_login_at	TIMESTAMP	Nullable	Date and time of the most recent successful login

Table 11:system-users data dictionary

Data dictionary for chw-profiles Table

Stores Community Health Worker specific profile data linked to the base user account.

Field	Data Type	Constraints	Description
profile_id	UUID	Primary Key	Unique identifier for the CHW profile
user_id	UUID	Foreign Key → system_users	Links to the CHW's authentication record
full_name	VARCHAR(100)	NOT NULL	Full name of the CHW
phone_number	VARCHAR(20)	NOT NULL	Primary contact

			number for field communication
assigned_village	VARCHAR(100)	NOT NULL	Village or sector the CHW is responsible for
facility_id	VARCHAR(50)	NOT NULL	Health center the CHW reports to
employee_id	VARCHAR(30)	Unique	Official government CHW identification number
Sync_status	ENUM	Default: SYNCED	Sync state for offline-created profile updates

Table 12:chw-profiles data dictionary

Data dictionary for clinical_staff_profiles Table

Stores facility provider and supervisor specific profile data.

Field	Data Type	Constraints	Description
profile_id	UUID	Primary Key	Unique identifier for the clinical staff profile
user_id	UUID	Foreign Key → system_users	Links to the staff member's authentication record
full_name	VARCHAR(100)	NOT NULL	Full name of the clinical staff member
facility_name	VARCHAR(100)	NOT NULL	Name of the health facility they are based at
specialization	VARCHAR(50)	Nullable	Clinical specialization such as HIV or TB
supervisor_level	BOOLEAN	Default: false	Whether this staff member has supervisor-level access
phone_number	VARCHAR(20)	NOT NULL	Contact number for alert notifications

Table 13:clinical-staff-profiles data dictionary

Data dictionary for patient_profiles Table

The core patient record holding demographic and clinical enrollment data.

Field	Data Type	Constraints	Description
patient_id	UUID	Primary Key	Unique identifier for each enrolled patient
user_id	UUID	Foreign Key → system_users	Links to the patient's authentication record
Full_name	VARCHAR(100)	NOT NULL	Full name of the patient
Date_of_birth	DATE	NOT NULL	Patient date of birth
sex	ENUM	NOT NULL	Patient sex: MALE, FEMALE, OTHER
diagnosis_type	ENUM	NOT NULL	Diagnosis: HIV, TB, HIV_TB_COINFECTION
art_start_date	DATE	Nullable	Date antiretroviral therapy was initiated
tb_start_date	DATE	Nullable	Date TB treatment was initiated
assigned_chw_id	UUID	Foreign Key → chw_profiles	The CHW responsible for this patient
phone_number	VARCHAR(20)	Nullable	Patient phone number for SMS notifications
notification_method	ENUM	Default: APP	Preferred notification channel: APP or SMS
fhir_patient_id	VARCHAR(100)	Nullable	FHIR Patient resource ID assigned after first sync
sync_status	ENUM	Default: PENDING	Sync state: PENDING, SYNCED, FAILED
created_at	TIMESTAMP	Auto-set	Date and time of patient registration

Table 14:patient-profiles data dictionary

Data dictionary for treatment_plans Table

Holds the medication regimen and dosing schedule prescribed for each patient.

Field	Data Type	Constraints	Description
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plan_id	UUID	Primary Key	Unique identifier for the treatment plan
patient_id	UUID	Foreign Key → patient_profiles	The patient this plan belongs to
medication_name	VARCHAR(100)	NOT NULL	Name of the prescribed medication or regimen
dosage	VARCHAR(50)	NOT NULL	Prescribed dose amount per administration
frequency	VARCHAR(50)	NOT NULL	Dosing schedule such as once daily or twice daily
start_date	DATE	NOT NULL	Date treatment was initiated
end_date	DATE	Nullable	Expected or actual end date of the treatment plan
isActive	BOOLEAN	Default: true	Whether this plan is currently in force
fhir_careplan_id	VARCHAR(100)	Nullable	FHIR CarePlan resource ID assigned after first sync
sync_status	ENUM	Default: PENDING	Sync state: PENDING, SYNCED, FAILED
created_by	UUID	Foreign Key → system_users	The clinical staff member who created the plan

Table 15: treatment-plans data dictionary

Data dictionary for home_visits Table

Records every CHW home visit including pill count, adherence observations, and clinical notes.

Field	Data Type	Constraints	Description
visit_id	UUID	Primary Key	Unique identifier for the home visit record
patient_id	UUID	Foreign Key → patient_profiles	The patient who was visited
chw_id	UUID	Foreign Key → chw_profiles	The CHW who conducted the visit
visit_date	DATE	NOT NULL	Date the home visit was conducted
adherence_status	ENUM	NOT NULL	Observed

			adherence: ADHERENT, NON_ADHERENT, PARTIAL
pill_count	INTEGER	NOT NULL	Actual number of pills remaining at time of visit
expected_pill_count	INTEGER	NOT NULL	Expected pills remaining based on confirmed doses
discrepancy_flagged	BOOLEAN	Default: false	Whether pill count contradicts confirmed dose history
symptoms	TEXT	Nullable	Patient-reported symptoms or side effects
notes	TEXT	Nullable	CHW general observations and clinical notes
Next_visit_date	DATE	NOT NULL	Scheduled date of the next home visit
fhir_observation_id	VARCHAR(100)	Nullable	FHIR Observation resource ID assigned after sync
sync_status	ENUM	Default: PENDING	Sync state: PENDING, SYNCED, FAILED
created_at	TIMESTAMP	Auto-set	Timestamp when the visit record was created

Table 16:home-visits data dictionary

Data dictionary for medication_records Table

Tracks the number of doses scheduled versus doses confirmed taken for each patient in each period, and stores the computed adherence percentage.

Field	Data Type	Constraints	Description
record_id	UUID	Primary Key	Unique identifier for the medication record
patient_id	UUID	Foreign Key → patient_profiles	The patient this record belongs to
plan_id	UUID	Foreign Key → treatment_plans	The treatment plan this record is tracking
period_start	DATE	NOT NULL	Start date of the tracking period
period_end	DATE	NOT NULL	End date of the

			tracking period
doses_scheduled	INTEGER	NOT NULL	Total doses expected in this period
doses_confirmed	INTEGER	NOT NULL	Total doses confirmed by patient
doses_verified	INTEGER	NOT NULL	Total doses verified by CHW pill count
adherence_pct	DECIMAL(5,2)	NOT NULL	Computed adherence percentage for the period
below_threshold	BOOLEAN	Default: false	Whether adherence has dropped below the alert threshold
false_confirmation_flag	BOOLEAN	Default: false	Whether AI has flagged a suspicious confirmation pattern
fhir_statement_id	VARCHAR(100)	Nullable	FHIR MedicationStatement resource ID after sync
sync_status	ENUM	Default: PENDING	Sync state: PENDING, SYNCED, FAILED
updated_at	TIMESTAMP	Auto-set	Timestamp of the most recent update

Table 17: medication-records data dictionary

Data dictionary for alerts Table

Stores all system-generated notifications including missed dose alerts, false confirmation flags, clinical outcome discrepancies, and population-level early warnings.

Field	Data Type	Constraints	Description
alert_id	UUID	Primary Key	Unique identifier for each alert record
patient_id	UUID	Foreign Key → patient_profiles	The patient this alert concerns
generated_for	UUID	Foreign Key → system_users	The user the alert is addressed to
alert_type	ENUM	NOT NULL	Type: MISSED_DOSE, FALSE_CONFIRMATION, CLINICAL_DISCREPANCY, EARLY_WARNING, STOCKOUT
severity	ENUM	NOT NULL	Severity level: INFO, WARNING, CRITICAL
title	VARCHAR(100)	NOT NULL	Short heading for the alert notification
message	TEXT	NOT NULL	Detailed description of the alert condition

is_read	BOOLEAN	Default: false	Whether the recipient has viewed the alert
is_resolved	BOOLEAN	Default: false	Whether the underlying condition has been addressed
resolved_by	UUID	Foreign Key → system_users	The user who resolved the alert
resolved_at	TIMESTAMP	Nullable	Timestamp when the alert was resolved
created_at	TIMESTAMP	Auto-set	Timestamp when the alert was generated

Table 18:alerts data dictionary

Data dictionary for stock_records Table

Tracks each CHW's current medication inventory with AI-generated days-remaining forecasts and automated resupply request status.

Field	Data Type	Constraints	Description
stock_id	UUID	Primary Key	Unique identifier for the stock record
chw_id	UUID	Foreign Key → chw_profiles	The CHW this stock belongs to
medication_name	VARCHAR(100)	NOT NULL	Name of the medication being tracked
current_quantity	INTEGER	NOT NULL	Current number of units in stock
unit	VARCHAR(20)	NOT NULL	Unit of measurement such as tablets or vials
reorder_level	INTEGER	NOT NULL	Quantity threshold that triggers a resupply request
days_remaining	INTEGER	Computed	AI forecast of days until stockout at current rate

resupply_requested	BOOLEAN	Default: false	Whether an automated resupply request has been submitted
resupply_requested_at	TIMESTAMP	Nullable	Timestamp when the resupply request was submitted
last_updated	TIMESTAMP	Auto-set	Timestamp of the most recent stock update
sync_status	ENUM	Default: PENDING	Sync state: PENDING, SYNCED, FAILED

Table 19:stock-records data dictionary

Data dictionary for dispensing_events Table

Records every individual medication dispensing transaction linking a CHW, a patient, and a specific stock record.

Field	Data Type	Constraints	Description
event_id	UUID	Primary Key	Unique identifier for the dispensing event
stock_id	UUID	Foreign Key → stock_records	The stock record that was decremented
patient_id	UUID	Foreign Key → patient_profiles	The patient who received the medication
chw_id	UUID	Foreign Key → chw_profiles	The CHW who dispensed the medication
medication_name	VARCHAR(100)	NOT NULL	Name of the medication dispensed

quantity_dispensed	INTEGER	NOT NULL	Number of units dispensed in this transaction
dispensed_at	TIMESTAMP	NOT NULL	Date and time of dispensing
notes	TEXT	Nullable	Any observations related to the dispensing event
sync_status	ENUM	Default: PENDING	Sync state: PENDING, SYNCED, FAILED

Table 20:dispensing-events data dictionary

Data dictionary for fhir_sync_logs Table

Maintains a complete audit log of every FHIR synchronization attempt including success and failure details.

Field	Data Type	Constraints	Description
log_id	UUID	Primary Key	Unique identifier for the sync log entry
initiated_by	UUID	Foreign Key → system_users	The user or device that triggered the sync
sync_direction	ENUM	NOT NULL	Direction: OUTBOUND (to server) or INBOUND (from EHR)
resource_type	VARCHAR(50)	NOT NULL	FHIR resource type: Patient, Observation, MedicationStatement, CarePlan
resource_id	VARCHAR(100)	NOT NULL	ID of the local record being synced
fhir_resource_id	VARCHAR(100)	Nullable	FHIR server-assigned resource ID on

			success
sync_status	ENUM	NOT NULL	Result: SUCCESS, FAILED, PARTIAL
error_message	TEXT	Nullable	Error details if the sync attempt failed
synced_at	TIMESTAMP	Auto-set	Timestamp when the sync was attempted

Table 21:*fhir_sync_logs data dictionary*

Data dictionary for ai_risk_scores Table

Stores the daily AI-generated adherence risk score and priority ranking for every active patient, maintaining a history that allows trend analysis over time.

Field	Data Type	Constraints	Description
score_id	UUID	Primary Key	Unique identifier for the risk score record
patient_id	UUID	Foreign Key → patient_profiles	The patient this score belongs to
score_date	DATE	NOT NULL	The date this score was calculated
risk_score	DECIMAL(5,2)	NOT NULL	Computed risk score from 0 (low risk) to 100 (critical)
risk_category	ENUM	NOT NULL	Category: LOW, MODERATE, HIGH, CRITICAL
priority_rank	INTEGER	NOT NULL	Rank within the CHW's patient list for this date
contributing_factors	JSONB	Nullable	JSON object listing the factors

			that drove the score
false_confirmation_signal	BOOLEAN	Default: false	Whether false confirmation pattern was detected today
clinical_discrepancy_signal	BOOLEAN	Default: false	Whether a lab result discrepancy was detected today
calculated_at	TIMESTAMP	Auto-set	Exact timestamp when the AI engine generated this score

Table 22:ai_risk_scores data dictionary

System Architecture Design

System architecture defines the overall structure of a system, describing how its components are organized, how they interact, and the principles guiding their design. It ensures key qualities such as scalability, reliability, and maintainability across the system (ISO/IEC/IEEE 42010, 2022).

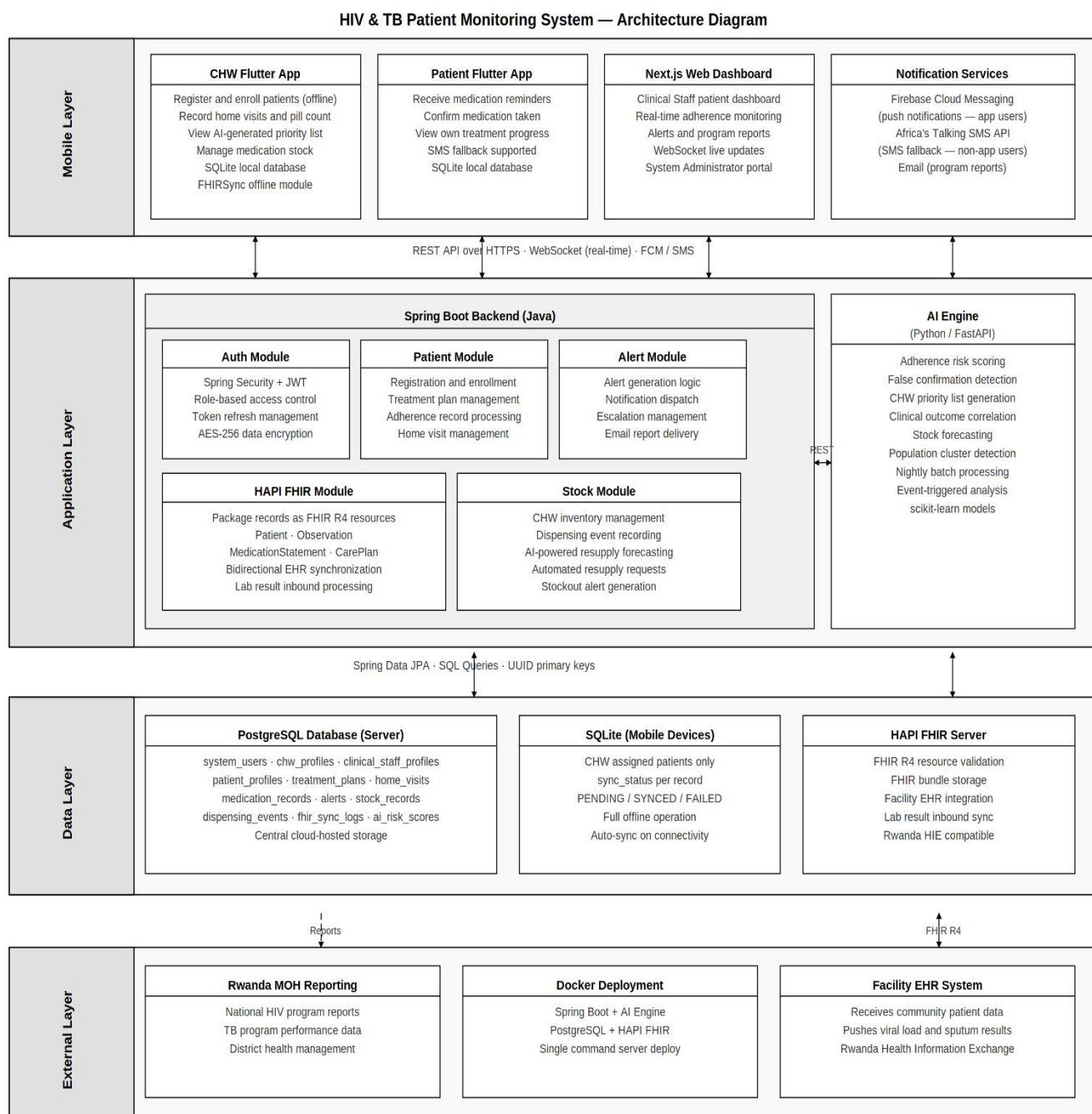


Figure 10: System Architecture Design

The HIV & TB Patient Monitoring System is organized into four layers:

- 1. Mobile Layer:** consists of two Flutter applications, one for Community Health Workers and one for Patients. Both apps operate fully offline using a local SQLite database and synchronize automatically with the server when connectivity is restored.

2. **Communication Layer:** handles all data exchange between the mobile apps, the web dashboard, and the backend. REST API calls over HTTPS handle standard requests while WebSocket connections deliver real-time alerts and dashboard updates to Clinical Staff without requiring page refresh.
3. **Application Layer:** runs on a Spring Boot backend hosted on a cloud server. This layer contains six modules: the Authentication Module managing JWT-based login and role-based access control, the Patient Management Module handling registration and treatment plans, the Adherence Module processing dose confirmations and visit records, the HAPI FHIR Module packaging and transmitting data as FHIR R4 resources, the Alert Module generating and dispatching notifications, and the AI Engine a Python FastAPI microservice called by Spring Boot to run risk scoring, false confirmation detection, priority list generation, clinical outcome correlation, stock forecasting, and cluster detection.
4. **Data Layer:** which consists of a central PostgreSQL database on the server storing all system data, and local SQLite databases on each mobile device storing only the data relevant to that specific user.

The four-layer design ensures that each layer can be updated or replaced independently without disrupting the others. The mobile layer can be updated without changing the backend. The AI engine can be retrained with new data without touching the Flutter apps. The FHIR integration layer can be reconfigured to connect to a different EHR system without affecting any other component.